MAU11E01 - Engineering Mathematics I
WEEK 1

Mathematical notation

- $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$ (the natural numbers)
- $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ (the integers)
- $\mathbb{R}$ (The set of real numbers - what you would most commonly think of as just 'numbers')

Notation

If a and b are real numbers, then..

- $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ is the *closed interval*.
- $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$ is the *open interval*.
- $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$ is the *half open interval*.
- $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$ is the *half open interval*.

An open end of an interval may assume infinite values, i.e. $[0, \infty)$ is all of the non-negative real numbers, where as $(-\infty, \infty)$ is just another notation for the whole real line, $\mathbb{R}$. We do not write $[-\infty, \infty]$ as we do not count " $\pm\infty$" as real numbers!

Functions

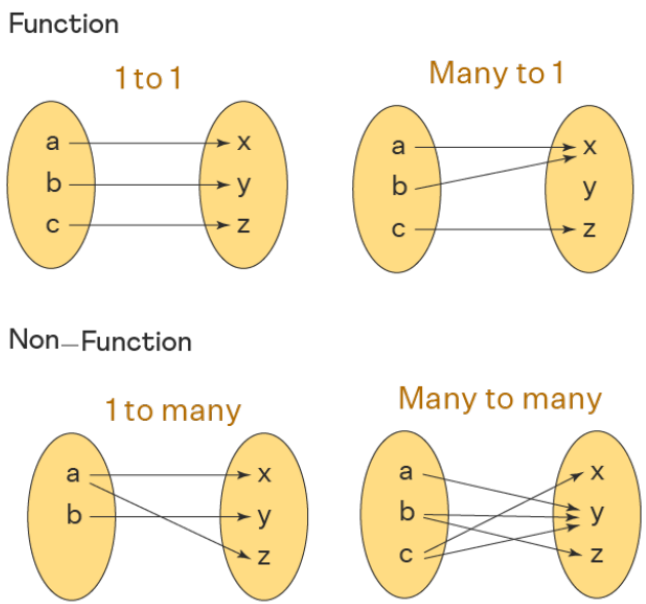
We start this course by studying the concept of a function.

Definition (A rough definition...)

We think of a function as a rule which takes an input and gives an output.

Definition

A function is an assignment of an input to an output such that every input is assigned to exactly one output.



How do we represent functions?

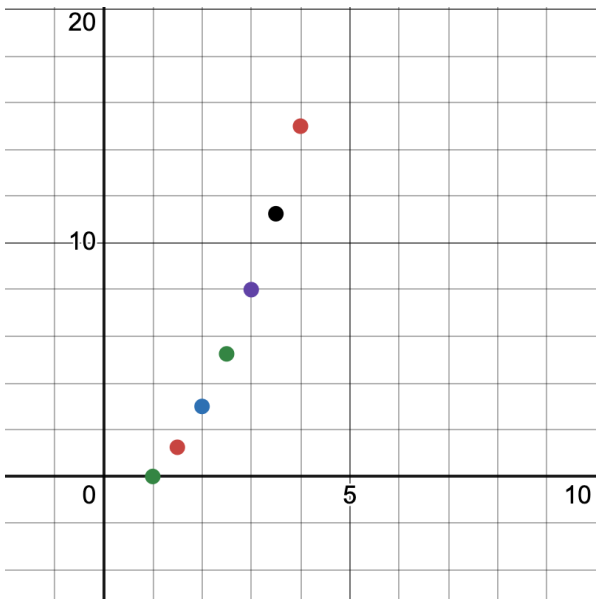
- A function can be represented by a table, for example

x	1	1.5	2	2.5	3	3.5	4
y	0	1.25	3	5.25	8	11.25	15

- A function can be represented by an algebraic formula, for example

$$y = x^2 - 1.$$

- A function can be represented by a graph, for example



How do we represent functions?

- A function can be represented verbally, for example

"to compute the value of y corresponding to a certain value, one should multiply that value by itself and subtract 1".

Changing symbols of a function

We could change the symbols used to denote the input and output;

$$f(x) = \frac{x-1}{2}, \qquad g(t) = \frac{t-1}{2}.$$

In both cases, f and g define the **same functions**, i.e. the rule for assignment of the input to the output is **the same**.

Example

- $f(x) =$
- $p(r) =$
- $Q(z) =$

How do we represent such functions?

Specific values of output of the function can be expressed symbolically, e.g. $f(2)$, is the output of f when the input is 2.

Exercise

If $f(x) = \dfrac{2x^2 - 4x + 3}{x - 2}$, calculate $f(-2), f(0)$ and $f(3)$.

Answer:

- $f(-2) =$
- $f(0) =$
- $f(3) =$

How do we represent such functions?

Vocabulary

We often use functions to describe how a quantity depends on another. We call the input of a function the independent variable and the output the dependent variable.

Function - (Slightly better)

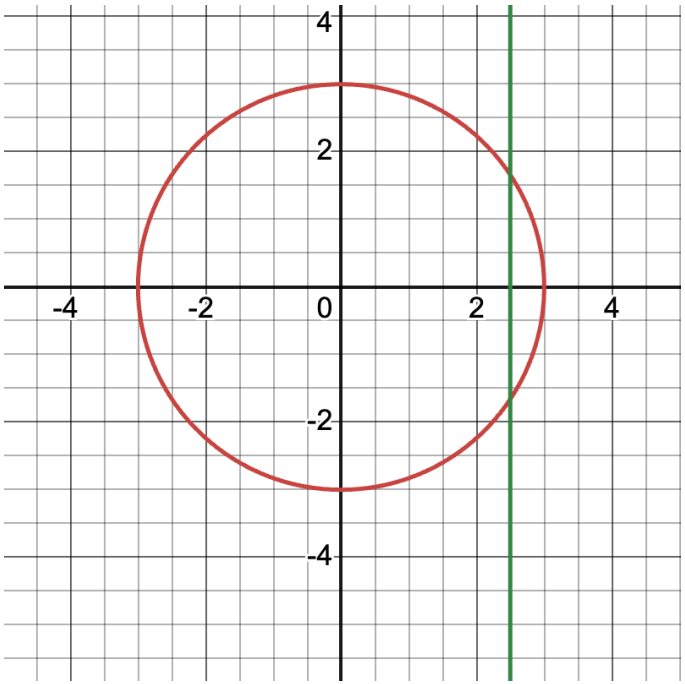
A (real valued) function f of a (real) variable x is a rule that associates a unique output with each input value of x. That output is denoted by $f(x)$.

Vertical Line Test

A curve in the xy-plane is the graph of some function exactly if no vertical line crosses it more than once.

Examples of the vertical line test

Consider the circle of the form $x^2 + y^2 = 9$ of radius 3. There are many vertical lines that intersect the circle twice:



- The vertical line test fails and so this circle is not a graph of a function.
- Algebraically, $y^2 = 9 - x^2$ means that $y = \pm\sqrt{9 - x^2}$, so the value of y can only be determined up to a sign, and hence it is not unique.

Examples of functions: The Square root

- A *square root* of a number x is defined to be a number y such that

$$y^2 = x.$$

- If y is a square root of x then so is $-y$ since

$$(-y)^2 = (-1)^2 y^2 = y^2 = x.$$

Hence for each $x > 0$ there are two square roots, a positive one, denoted by $\sqrt{x}$ and a negative one, namely $-\sqrt{x}$.

- For $x = 0$, we have $\sqrt{0} = 0$, and for $x < 0$, there is no real number y such that $y^2 = x$, i.e. no square root of x when $x < 0$.

Definition

The square root function is defined as the function $f(x) = \sqrt{x}$, where $x \geq 0$.

This is an example of a function which

- Is defined only for $x \geq 0$, i.e. not for all x values
- Only returns non-negative values.

Examples of functions: Absolute value

- The absolute value or magnitude of a real number x is defined by

$$|x| = \sqrt{x^2} = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Properties of the absolute value

Let $a, b \in \mathbb{R}$. Then

i. $|-a| = a$

ii. $|a \cdot b| = |a| \cdot |b|$

iii. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ for $b \neq 0$.

iv. $|a + b| \leq |a| + |b|$ (triangle-inequality)

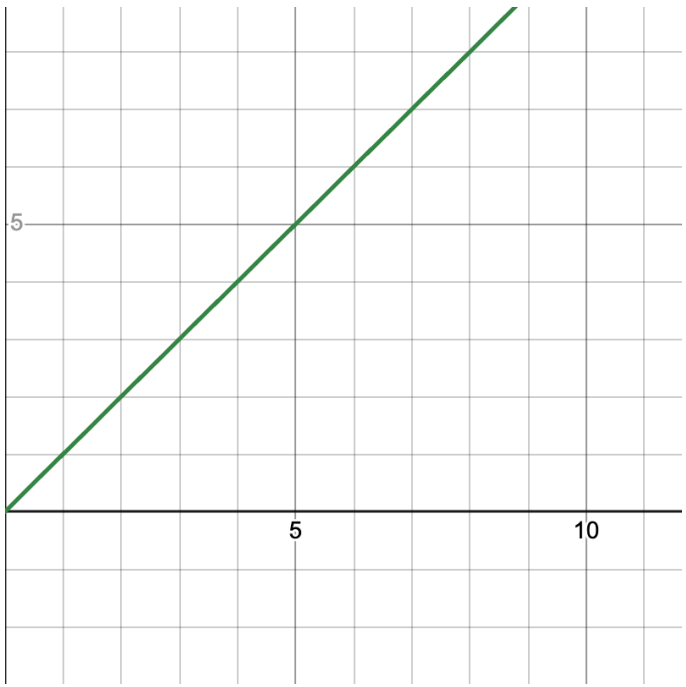
- Students often write $\sqrt{x^2} = x$, which is false in general!
- The absolute value is an example of a piecewise defined function.

Piecewise defined functions

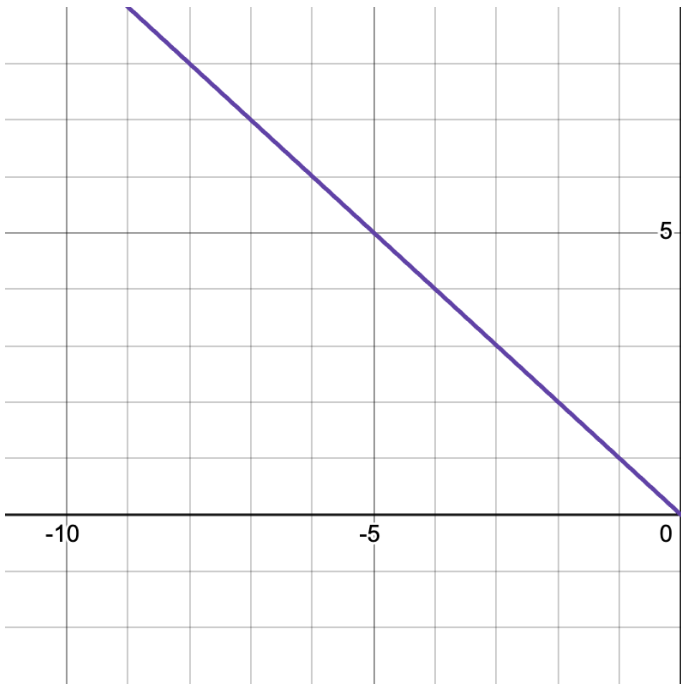
Let $g(x) = x \quad$ for $x \geq 0$, $h(x) = -x \quad$ for $x < 0$, and

$$f(x) = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases},$$

and consider the following graphs:



(a) $g(x) = x$



(b) $h(x) = -x$

Piecewise defined functions

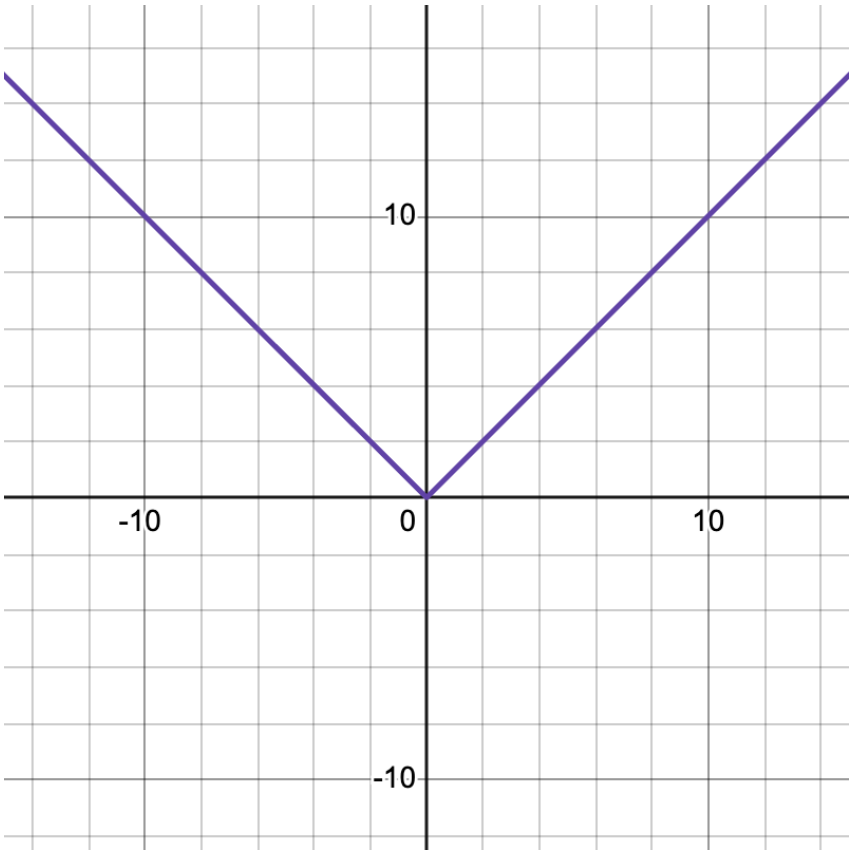


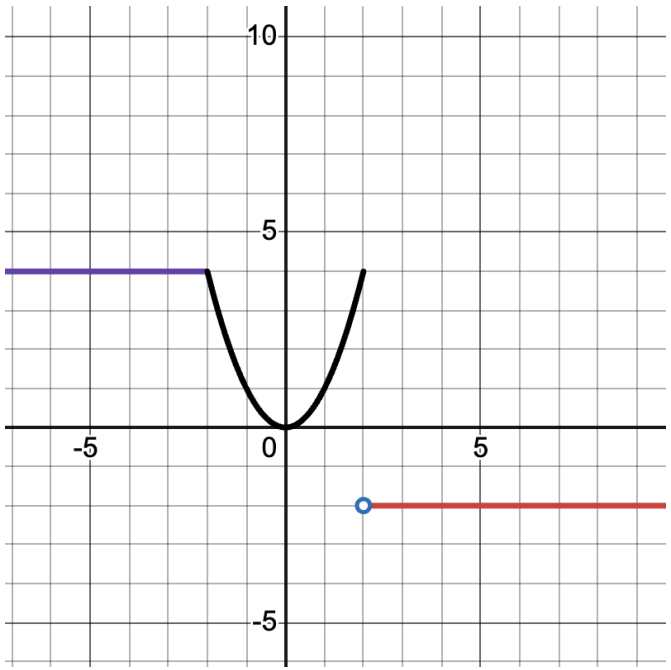
Figure: The graph of $f(x) = |x|$.

Piecewise defined functions

As an example, let's consider the function

$$f(x) = \begin{cases} 4, & x \leq -2 \\ x^2, & -2 < x \leq 2 \\ -2, & x > 2 \end{cases}$$

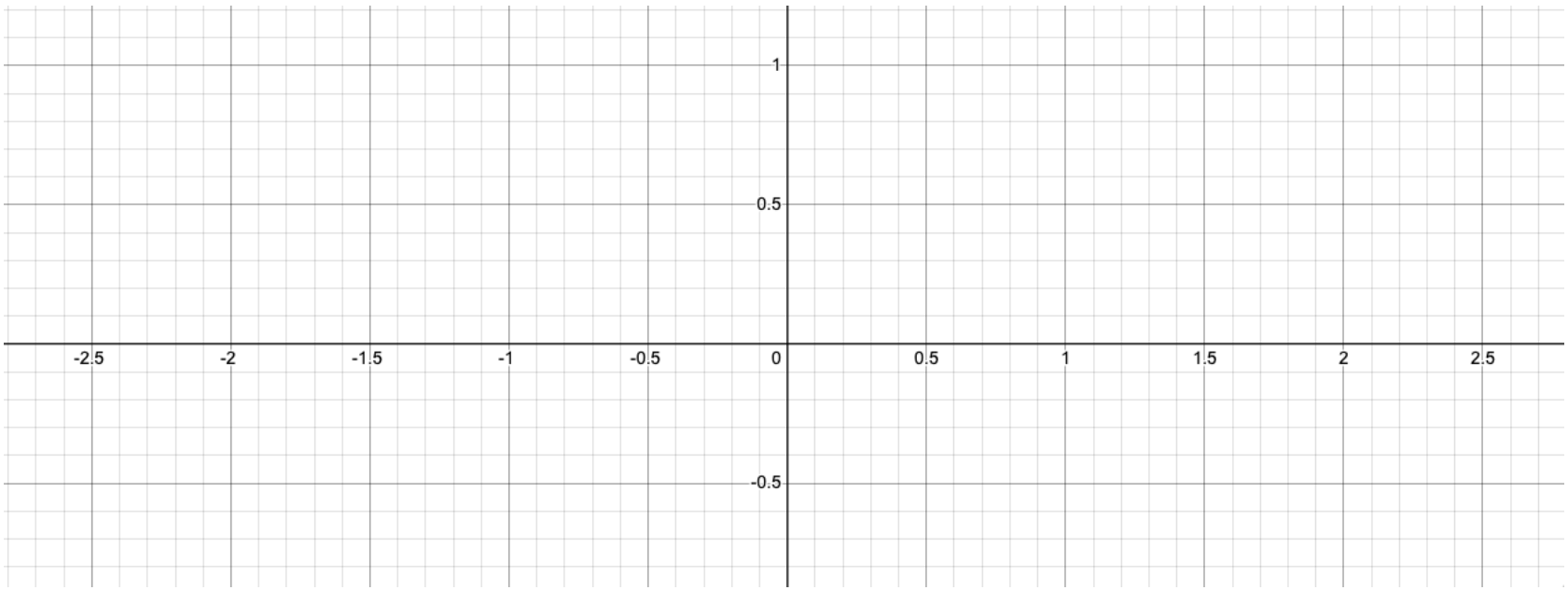
- At the value $x = -2$ the function is continuous.
- At the value $x = 2$ the function makes a jump.
- We say that the function f is discontinuous at the point $x = 2$. (We will come to this later)



Practice with piecewise functions

Let's try a more complicated example. Plot the function

$$g(x) = \begin{cases} 0, & x \leq -2, \\ x + 2, & -2 < x \leq -1, \\ 1 & -1 < x \leq 1, \\ -x^2 + 2, & x > 1. \end{cases}$$



The domain and range of functions

Domain of a function

The domain of a function f is the set of all real values x for which f is to take as an input, and $f(x)$ is a real valued output.

Range of a function

The range of a function is the set of values $f(x)$ where x is in the domain of f.

Sometimes we will *declare* the domain of a real valued function by

$$f : A \to \mathbb{R},$$

and then by some formula $f(x) = ??$

- A is the set of real number in which f is to take its inputs from. e.g. $A = (0, 1)$, $[-3, 10]$, $[0, \infty)$ etc.
- The $\mathbb{R}$ here is just to say that the outputs will take real values. $\mathbb{R}$ is not necessarily the range of f!

Lets look at some examples.

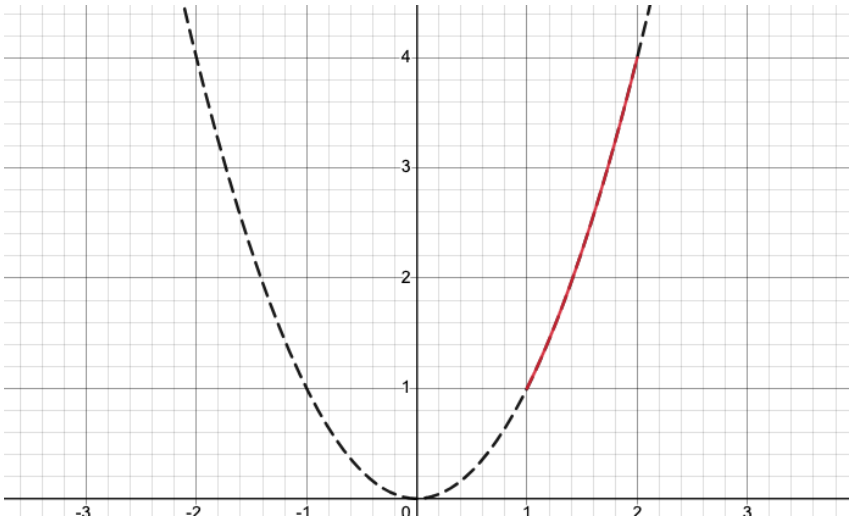
The domain and range of functions

Example

Consider the function $f : [1, 2] \to \mathbb{R}$ defined by

$$f(x) = x^2$$

Domain of f = Range of f =



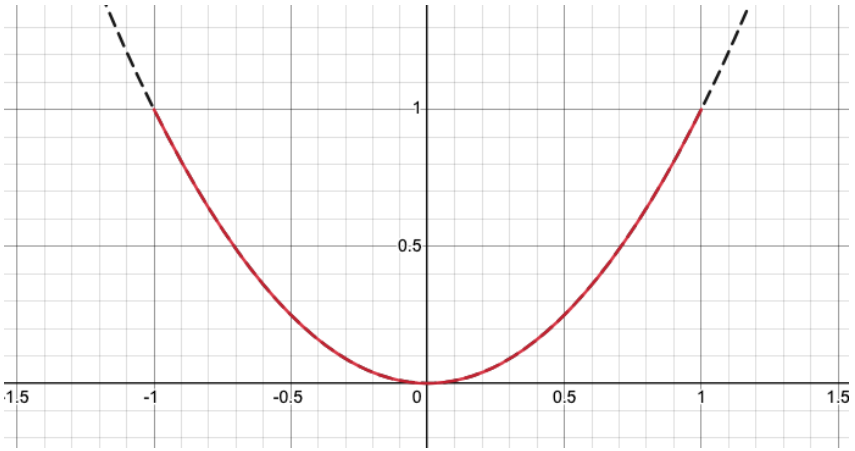
The domain and range of functions

Example

Consider the function $f : [-1, 1] \to \mathbb{R}$ defined by

$$f(x) = x^2$$

Domain of f = Range of f =



The domain and range of functions

Sometimes we will declare functions without a domain and just by a formula.

Natural domain of a function

The natural domain of a function f defined only by a formula, consists of all (real) values of x such that $f(x)$ has a well defined real value.

Example

Consider the square root function $f(x) = \sqrt{x}$.

Domain of f = Range of f =



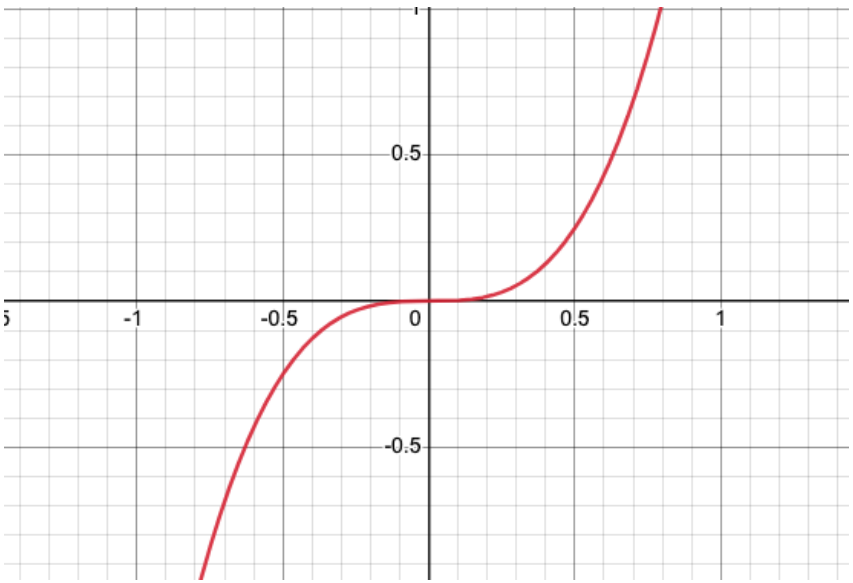
The domain and range of functions

Example

Consider the function $g(x) = x^3$.

Domain of g =

Range of g=

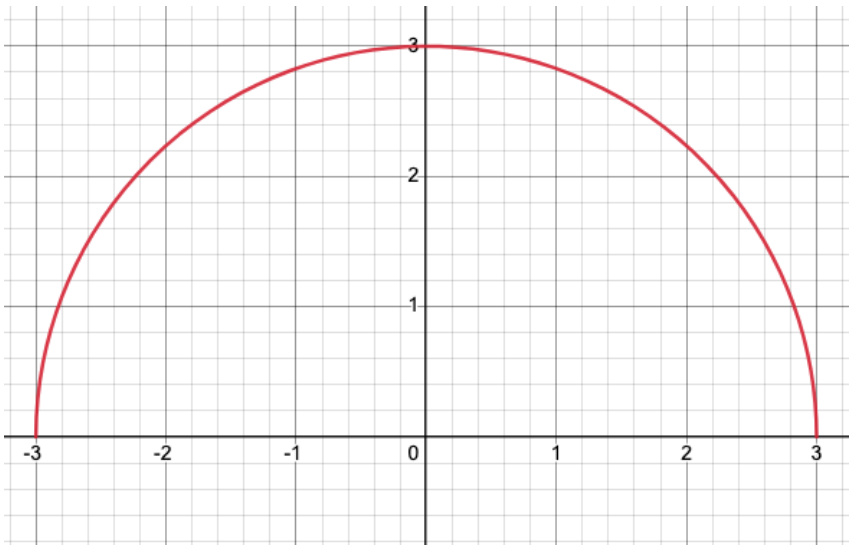


The domain and range of functions

Example

Consider the function $y = \sqrt{9 - x^2}$.

Domain of $y =$ Range of $y=$



Exercise

Can you find the domain and range for the function $y = -\sqrt{25 - x^2}$?

The domain and range of functions

Example

Consider the function $h(x) = x^2 + 2x - 3$.

Domain of $h =$ Range of $h=$

It might be helpful to *complete the square;*

$h(x) =$

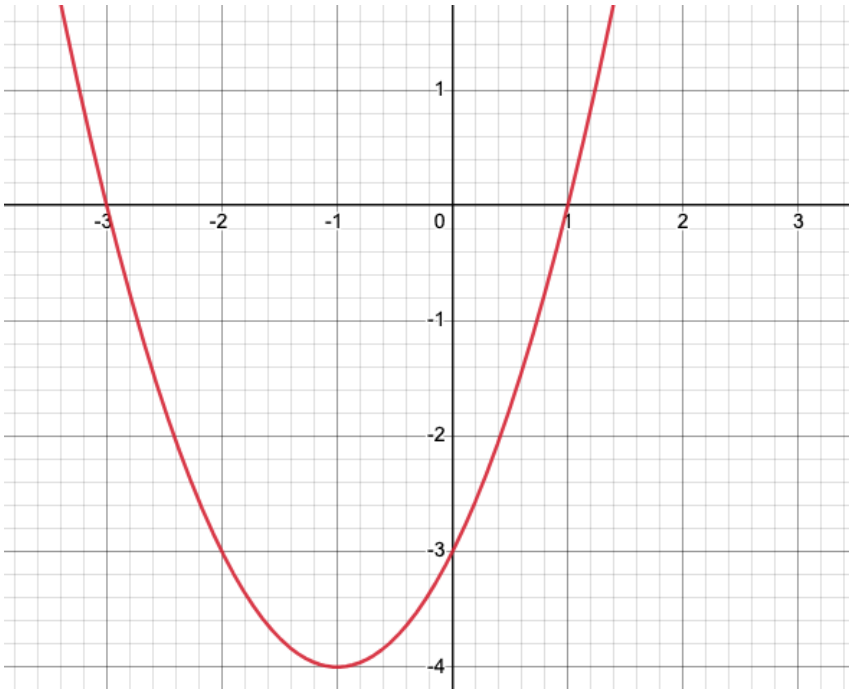
The domain and range of functions

Example

Consider the function $h(x) = x^2 + 2x - 3 = (x+1)^2 - 4$.

Domain of h =

Range of h=



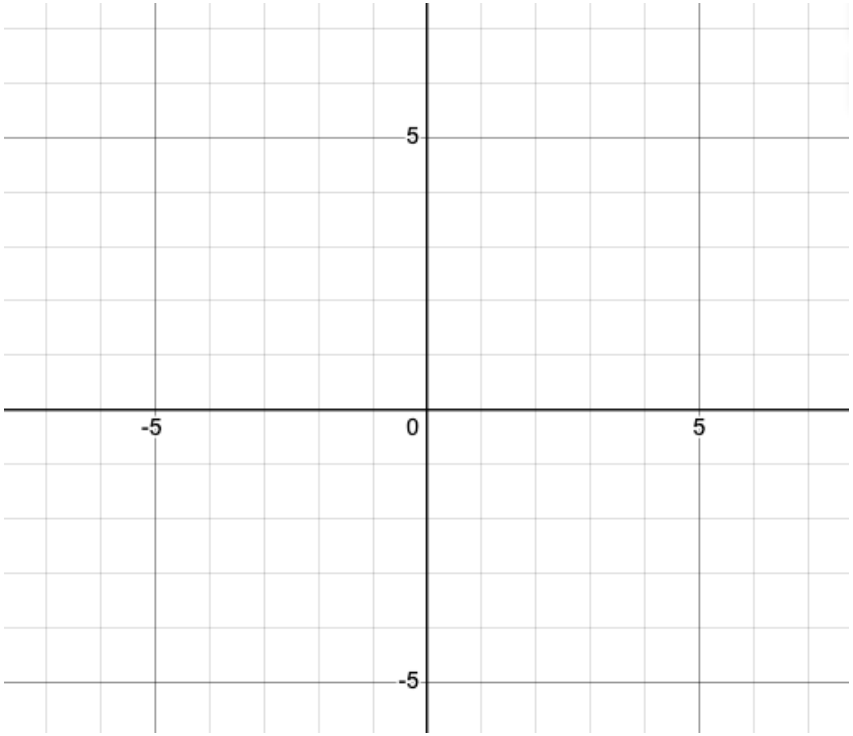
The domain and range of functions

Example

Consider the function $p(x) = \frac{1}{(x-1)(x+1)}$.

Domain of p =

Range of p=



The domain and range of functions

After all the examples above, the following algorithm for describing the natural domain of a basic function defined by a formula should appear intuitive to you:

1) Whenever you see an expression of type $\sqrt{g(x)}$ (or $\sqrt[2n]{g(x)}$ with $n \in \mathbb{N}$), determine all x for which $g(x) \geq 0$.

2) Whenever you see an expression of type $\frac{h(x)}{g(x)}$, determine all x for which $g(x) \neq 0$.

Beware of the functions $\tan(x) = \frac{\sin(x)}{\cos(x)}$ and $\cot(x) = \frac{\cos(x)}{\sin(x)}$, which are implicit fractions!

3) Determine the intersection of all the parts of $\mathbb{R}$ obtained in the previous points.

The domain and range of functions

Example

The range of $f(x) = 2 + \sqrt{x-1}$ is $[2, +\infty)$.

To see that, we observe that the natural domain of this function is $[1, +\infty)$ since we request that the expression from which we extract the square root is non-negative.

Now, as x increases from 1 to $+\infty$,

- $x - 1$ varies in $[0, +\infty)$,
- $\sqrt{x-1}$ varies in $[0, +\infty)$,
- and $2 + \sqrt{x-1}$ varies in $[2, +\infty)$.

The domain and range of functions

Example

The range of $f(x) = \dfrac{x+1}{x-1}$ is harder to guess, so we shall approach it directly from the definition. We would like to find the values of a for which the equation $\dfrac{x+1}{x-1} = a$ has solutions. Solving it, we get

$$\begin{aligned} x+1 &= ax-a, \\ a+1 &= ax-x = x(a-1), \\ x &= \frac{a+1}{a-1}. \end{aligned}$$

From $x = \dfrac{a+1}{a-1}$, we see that all $a \in \mathbb{R}$ except for $a = 1$ are in the natural range. So, the natural range is $(-\infty, 1) \cup (1, +\infty)$.

New Functions from Old

RECALL

- The intersection of A and B consists of those points that belong to both A and B :
$$A \cap B = \{x \in \mathbb{R} \mid x \in A \text{ and } x \in B\}.$$
- The union of A and B consists of those points that belong either to A or else to B :
$$A \cup B = \{x \in \mathbb{R} \mid x \in A \text{ or } x \in B\}.$$

We can combine function by performing arithmetic with them;

Definition

Given two functions f and g, and for $x \in \mathbb{R}$ we define

- $(f+g)(x) = f(x) + g(x)$,
- $(f-g)(x) = f(x) - g(x)$,
- $(fg)(x) = f(x) \cdot g(x)$,
- $(f/g)(x) = \frac{f(x)}{g(x)}$.

The domains of the above functions are given by the intersection of the two domains - except from f/g where we must be careful to exclude the

New Functions from Old

Example

If $f : [-10, 10] \to \mathbb{R}$ and $g : [-10, 0] \to \mathbb{R}$ are defined respectively by

$$f(x) = x + 2x^2, \qquad g(x) = x + 1,$$

then $f + g : [-10, 0] \to \mathbb{R}$ is defined by $(f + g)(x) = 1 + 2x + 2x^2$.

Example

If $f : (0, 5) \to \mathbb{R}$ and $g : (2, \infty) \to \mathbb{R}$ are defined respectively by

$$f(x) = \sin x, \qquad g(x) = x^2 - 1,$$

then $f/g : A \to \mathbb{R}$ is defined by $(f/g)(x) = \dfrac{\sin x}{x^2 - 1}$, where

$$\begin{aligned} A &= \{x \in \mathbb{R} \mid 2 < x < 5 \text{ and } x \neq -1, 1\} \\ &= \{x \in \mathbb{R} \mid 2 < x < 5\} \end{aligned}$$

New Functions from Old

Example

Let $f(x) = 1 + \sqrt{x-2}$, and $g(x) = x - 3$. Then

- $(f+g)(x) = 1 + \sqrt{x-2} + x - 3 = x - 2 + \sqrt{x-2}$,
- $(f-g)(x) = 1 + \sqrt{x-2} - (x-3) = 4 - x + \sqrt{x-2}$,
- $(fg)(x) = (1 + \sqrt{x-2})(x-3)$.

In all these cases, the domain is the intersection of the domains of f and g, which is given by $[2, +\infty) \cap \mathbb{R} = [2, +\infty)$.

- $(f/g)(x) = \dfrac{1 + \sqrt{x-2}}{x-3}$.

Here the domain is $[2, +\infty) \setminus \{3\} = [2, 3) \cup (3, +\infty)$.

New Functions from Old

Example

Let $f(x) = \sqrt{x-2}$, and $g(x) = \sqrt{x-3}$. Then

$$(fg)(x) = \sqrt{x-2}\sqrt{x-3} = \sqrt{(x-2)(x-3)} = \sqrt{x^2-5x+6}.$$

The domain of fg is $[2,+\infty) \cap [3,+\infty) = [3,+\infty)$.

The natural domain of $\sqrt{x^2-5x+6}$ is $(-\infty,2] \cup [3,+\infty)$, since $x^2-5x+6 = (x-2)(x-3)$. This is different from $[3,+\infty)$!

Example

Let $f(x) = x$, and $g(x) = 1/x$. Then

$$(f/g)(x) = \frac{x}{\frac{1}{x}} = x^2.$$

The domain of f/g is $(-\infty,0) \cup (0,+\infty)$, which *does not* coincide with the natural domain of x^2, that is $(-\infty,+\infty)$.